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Dynamic geometric constraints govern covalent feasibility in KRAS G12C ligands

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Abstract

Covalent inhibition has emerged as an effective strategy for targeting oncogenic proteins, yet successful covalent engagement depends not only on binding affinity but also on the dynamic preservation of productive electrophile–thiol geometry. Motivated by the clinical success of acrylamide-based KRAS G12C inhibitors such as sotorasib, we performed a structure-based and molecular dynamics investigation of acrylamide-containing KRAS G12C ligands. Although several ligands initially achieved near-reactive proximity to Cys12, production molecular dynamics simulations revealed rapid relaxation of the pre-reactive geometry and loss of productive electrophile–thiol alignment. Similar geometric relaxation was observed for noncovalent sotorasib simulations, whereas covalent KRAS G12C complexes preserved stable sulfur–carbon distances and chemically consistent bond geometries throughout explicit-solvent molecular dynamics. These findings suggest that covalent feasibility in KRAS G12C is governed by stringent dynamic geometric constraints and emphasize the importance of reactive-state preorganization in covalent inhibitor design.

Keywords Covalent inhibitor design, KRAS G12C, Reactive geometry, Molecular dynamics simulation

1 Introduction

Covalent inhibition has re-emerged as a major strategy in modern drug discovery, enabling prolonged target engagement and enhanced selectivity through irreversible or quasi-irreversible interactions with nucleophilic residues [1, 2]. Recent clinical successes, including the KRAS G12C inhibitors sotorasib and adagrasib, have demonstrated that selective covalent targeting of cysteine residues can be achieved through precise structural and mechanistic optimization [3–6]. In KRAS G12C, these inhibitors exploit a transient Switch-II pocket in the inactive GDP-bound state to achieve productive engagement of CYS12, establishing covalent targeting as a clinically validated strategy for a previously difficult oncogenic target [3, 4].

Among currently developed KRAS G12C inhibitors, sotorasib and adagrasib represent the first clinically approved covalent agents and have established the feasibility of selectively targeting the mutant cysteine within the Switch-II pocket. Several additional



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KRAS G12C inhibitors, including divarasil, olomorasib, and other next-generation covalent inhibitors, are currently undergoing clinical or preclinical evaluation with the aim of improving potency, selectivity, resistance profiles, and pharmacokinetic properties. These advances highlight the continuing importance of understanding the structural and dynamic determinants governing productive covalent engagement.

KRAS is one of the most frequently mutated oncogenes in human cancer and plays a central role in regulating cellular proliferation, differentiation, and survival through the RAS–MAPK signaling pathway. Oncogenic mutations at residues such as G12, G13, and Q61 impair intrinsic GTP hydrolysis, resulting in constitutive activation of downstream signaling pathways that promote uncontrolled cell growth and tumor progression [7]. Among these variants, KRAS G12C is unique because substitution of glycine with cysteine introduces a nucleophilic residue that can be selectively targeted by electrophilic small molecules, transforming a historically undruggable target into a clinically actionable therapeutic opportunity. Recent computational and structural studies have further highlighted the conformational dynamics, molecular recognition mechanisms, and ligand-binding characteristics of KRAS, providing valuable insights for the rational development of next-generation covalent inhibitors [8, 9].

Successful covalent inhibition requires more than favorable binding affinity alone. Productive bond formation depends on the persistent alignment of an electrophilic warhead with a nucleophilic residue along a chemically favorable attack trajectory, commonly described in terms of electrophile–nucleophile distance and Bürgi–Dunitz geometry [10, 11]. Previous structure-based design studies have primarily focused on optimizing non-covalent interactions and docking-derived binding poses; however, considerably less attention has been given to whether near-reactive geometries remain dynamically stable under physiological conditions [12, 13]. Consequently, ligands that initially appear compatible in docking studies may fail to preserve productive electrophile–thiol alignment during molecular motion, limiting the formation of stable covalent interactions.

KRAS G12C presents a particularly stringent environment for covalent inhibitor design because productive engagement of CYS12 requires precise geometric preorganization within a conformationally dynamic binding pocket [3, 4]. Although acrylamide-based inhibitors such as sotorasib have demonstrated that clinically effective covalent targeting is achievable, it remains unclear to what extent alternative scaffolds can reproduce the geometric and conformational features required for productive covalent engagement. In particular, the relationship between apparent near-reactive docking proximity and dynamically persistent reactive configurations has not been systematically examined in KRAS G12C systems using comparative non-covalent and covalent molecular dynamics frameworks.

In the present study, we investigated covalent feasibility in KRAS G12C using an integrated workflow combining molecular docking, geometric analysis, and explicit-solvent molecular dynamics simulations. Initial scaffold evaluation identified several acrylamide-containing ligands capable of achieving near-reactive proximity to CYS12 during docking; however, subsequent molecular dynamics simulations revealed rapid relaxation of productive electrophile–thiol geometries. To further examine the structural basis of covalent accessibility, a geometry-constrained ligand configuration was manually reconstructed using reactive alignment features derived from the experimentally validated sotorasib binding mode and refined through iterative structural insights obtained across

multiple scaffold and ligand design attempts. Comparative simulations of non-covalent and explicitly covalent KRAS G12C complexes were then used to evaluate the dynamic persistence of reactive geometries during molecular motion.

Unlike conventional docking-centered workflows that prioritize affinity-driven pose selection, the present work frames ligand evaluation as a problem of covalent feasibility governed by dynamic geometric persistence. By directly comparing scaffold-derived ligands, manually reconstructed geometries, and benchmark sotorasib systems under both non-covalent and covalent simulation conditions, this study distinguishes transient reactive proximity from dynamically stable covalent accessibility. Accordingly, the present study investigates whether dynamic preservation of chemically favorable sulfur–electrophile geometry, rather than static docking proximity alone, provides a more reliable indicator of covalent feasibility in KRAS G12C inhibitor design. By integrating structure-based docking with explicit-solvent molecular dynamics simulations, the study aims to evaluate the dynamic geometric constraints governing productive covalent engagement and their implications for the rational design of both covalent and non-covalent KRAS inhibitors.

2 Computational methods

2.1 System preparation and ligand design

The KRAS G12C crystal structure (PDB ID: 5V9O), originally reported by Zeng et al. [14], was retrieved from the Protein Data Bank (PDB) and prepared by isolating chain A, removing non-essential components, and defining Cys12 as the reactive residue. Hydrogen atoms were added, and the resulting structure was prepared for docking and molecular dynamics simulations using OpenMM- and AmberTools-compatible workflows [15]. The experimentally validated KRAS G12C inhibitor sotorasib was additionally used as a benchmark reference for evaluating reactive geometry and covalent alignment features.

Ligands were constructed using RDKit through SMILES-based molecular enumeration, with all candidates incorporating an acrylamide electrophilic warhead consistent with clinically validated KRAS G12C inhibitors [16]. Ligand generation was performed by systematically varying aromatic scaffolds, linker architectures, and substituent patterns while preserving the conserved acrylamide electrophile and chemically reasonable molecular connectivity. Scaffold variations were introduced through iterative design rounds to explore substituent effects, linker modifications, and degrees of scaffold rigidification while maintaining compatibility with the KRAS Switch-II pocket. Three-dimensional conformers were subsequently generated using the ETKDG algorithm and optimized using the MMFF94 or UFF force fields prior to docking. Following each round of molecular docking, candidate ligands were evaluated based on their sulfur–electrophile distance, Bürgi–Dunitz attack angle, and overall electrophilic warhead orientation relative to Cys12. Structural insights obtained from one design cycle guided subsequent scaffold modifications, resulting in progressive refinement of electrophile positioning and pre-reactive geometry throughout the iterative ligand design process.

During iterative scaffold evaluation, selected ligand configurations exhibiting favorable reactive alignment were further refined through geometry-guided reconstruction **based on** the experimentally validated sotorasib binding mode. This process enabled preservation of chemically consistent sulfur–electrophile orientation features identified during

comparative geometric analysis and served as the basis for subsequent covalent molecular dynamics simulations.

2.2 Docking and geometric evaluation

Non-covalent docking was performed using AutoDock Vina with a grid centered on the KRAS Switch-II pocket [17]. Initial broad conformational sampling was followed by refinement using a reduced grid focused on the cysteine-adjacent region to improve evaluation of electrophile positioning relative to CYS12. Multiple docking poses were generated for each ligand to minimize pose-dependent bias and to evaluate the consistency of observed geometric trends across independent binding conformations.

Covalent docking was performed using AutoDock-GPU in reactive docking mode, enabling explicit modeling of covalent bond formation with CYS12 [18]. Multiple docking runs were conducted per ligand, and top-ranked poses were selected based on predicted affinity, electrophile accessibility, and geometric compatibility with the reactive cysteine residue. Benchmark sotorasib systems were evaluated under identical docking conditions to ensure methodological consistency during comparative analysis.

Covalent feasibility was assessed using two primary geometric criteria: (i) the distance between the electrophilic carbon and the sulfur atom of CYS12, and (ii) the nucleophilic attack geometry defined by the $SG-C\beta-C=O$ angle, consistent with Bürgi–Dunitz trajectory requirements for productive nucleophilic attack [10, 11]. Docked conformations were further evaluated for maintenance of chemically favorable sulfur–electrophile orientation and overall compatibility with productive pre-reactive alignment prior to molecular dynamics simulations.

2.3 Geometry guided covalent reconstruction

To further investigate the structural determinants of covalent accessibility, selected ligand configurations were refined through a geometry-guided reconstruction strategy based on reactive alignment features observed in the experimentally validated sotorasib–KRAS G12C complex. Initial non-covalent docking studies identified several ligands capable of achieving near-reactive proximity to CYS12; however, many of these configurations failed to preserve productive sulfur–electrophile geometry during subsequent molecular dynamics simulations.

To examine whether preservation of benchmark-derived reactive alignment could improve geometric stability, a manually reconstructed covalent ligand configuration was generated using a geometry-constrained core derived from the sotorasib binding paradigm and refined through iterative scaffold design observations obtained during earlier docking and simulation analyses. Particular emphasis was placed on maintaining chemically favorable sulfur–electrophile distance and Bürgi–Dunitz-compatible attack orientation while preserving compatibility with the KRAS Switch-II pocket environment.

The resulting reconstructed configuration was subsequently used for explicitly covalent molecular dynamics simulations and comparative evaluation against corresponding non-covalent systems to assess the dynamic persistence of productive reactive geometries under explicit-solvent conditions.

2.4 Molecular dynamics simulations

Initial non-covalent molecular dynamics simulations were performed to evaluate whether docked ligand configurations could maintain productive pre-reactive geometry under explicit-solvent conditions. Exploratory simulations of selected KRAS G12C ligand complexes, including the R18_A4 system, were conducted using OpenMM with the AMBER ff14SB force field applied to the protein and GAFF2-derived parameters assigned to ligand structures. Ligand topologies were generated through ACPYPE-compatible parameterization workflows, and complexes were solvated in explicit TIP3P water boxes with appropriate counterions added to maintain charge neutrality.

Following energy minimization, systems were gradually heated and equilibrated under near-physiological temperature and pressure conditions prior to initiating unrestrained production simulations. Long-range electrostatic interactions were treated using the particle mesh Ewald method, while bonds involving hydrogen atoms were constrained to permit a 2 fs integration time-step. Non-covalent production trajectories were subsequently analyzed to evaluate the stability of sulfur–electrophile proximity and the persistence of productive Bürgi–Dunitz-compatible attack geometry during molecular motion.

These simulations served primarily as dynamic feasibility tests to determine whether ligand configurations that appeared near-reactive during docking could maintain chemically favorable alignment in the absence of explicit covalent stabilization. Comparative non-covalent simulations were additionally performed for benchmark sotorasib systems to evaluate whether experimentally derived reactive geometries remained dynamically persistent under equivalent non-covalent conditions.

2.5 Covalent molecular dynamics simulations

For covalent molecular dynamics simulations, the KRAS G12C–ligand complex was parameterized within the AMBER force field framework. The protein was described using the ff14SB force field, whereas the ligand was parameterized using the General AMBER Force Field (GAFF). Ligand atom types and AM1-BCC partial atomic charges were assigned during generation of the ligand MOL2 file (lig.mol2), and any missing ligand parameters were supplied using the corresponding force-modification file (lig.frcmod). The protein and ligand were loaded separately into LEaP and combined into a single simulation system. A covalent bond was then explicitly introduced between the sulfur atom (SG) of Cys12 and the electrophilic carbon atom (C25) of the ligand using the LEaP bond command, after removal of the cysteine thiol hydrogen (HG) to represent the covalently modified state. Because this sulfur–carbon linkage is not included in the standard AMBER force fields, an additional parameter modification file (covalent.frcmod) was constructed to define the bonded interactions associated with the newly formed junction. This file introduced the equilibrium SG–C25 bond together with the corresponding bond force constant, adjacent bond-angle parameters, and torsional terms required to describe the covalent linkage, while all remaining bonded and non-bonded interactions were retained from the standard ff14SB and GAFF force fields. Following incorporation of these parameters, LEaP regenerated the complete molecular topology, and the resulting covalent complex was solvated in an explicit cubic TIP3P water box extending 10 Å beyond the solute in all directions. The final topology (complex_covalent.prmtop) and coordinate (complex_covalent.inpcrd) files generated by

LEaP were subsequently used for energy minimization, heating, equilibration, and production molecular dynamics simulations.

Long-range electrostatic interactions were treated using the particle mesh Ewald method, while bonds involving hydrogen atoms were constrained using the SHAKE algorithm, enabling a 2 fs integration timestep. Production trajectories were extended through sequential restart simulations using checkpoint structures from preceding runs, allowing continuous monitoring of structural stability and reactive geometry over progressively longer simulation timescales.

Both covalent sotorasib benchmark systems and geometry-guided reconstructed ligand complexes were evaluated using this workflow. These simulations were designed to determine whether explicit covalent stabilization could preserve productive sulfur–electrophile geometry that was not maintained during corresponding non-covalent simulations. Additional details regarding covalent ligand parameterization, custom force-field definitions, and topology generation are provided in the Supporting Information.

2.6 Trajectory and geometric analysis

Trajectory analysis was performed to evaluate the structural stability and dynamic persistence of pre-reactive geometries throughout molecular dynamics simulations. Particular emphasis was placed on monitoring the distance between the sulfur atom of CYS12 and the electrophilic carbon of the ligand warhead, together with the corresponding SG–C β –C = O nucleophilic attack geometry associated with productive Bürgi–Dunitz alignment [10, 11]. Distance and angle measurements were extracted from simulation trajectories using cpptraj-based analyses and subsequently evaluated across heating, equilibration, and production phases.

Comparative analyses were performed for both non-covalent and explicitly covalent KRAS G12C systems to distinguish transient near-reactive proximity from dynamically persistent covalent accessibility. Structural stability during simulation was additionally assessed through trajectory-based RMSD evaluation and qualitative inspection of ligand positioning within the Switch-II pocket. Average sulfur–electrophile distances, geometric fluctuations, and persistence of productive reactive alignment were used as primary indicators of covalent feasibility under explicit-solvent conditions.

3 Results and discussion

The computational workflow used in this study is summarized in Fig. 1. Starting from acrylamide-based scaffold design, ligands were evaluated through sequential docking, geometric filtering, and molecular dynamics simulations to determine whether apparent near-reactive configurations could maintain productive electrophile–thiol alignment under dynamic conditions. In contrast to conventional affinity-centered workflows, the present approach incorporated both geometric criteria and trajectory-level stability analysis to evaluate covalent feasibility in KRAS G12C systems [10–13]. Initial docking and geometric filtering identified several ligand poses with favorable sulfur–electrophile proximity; however, subsequent non-covalent molecular dynamics simulations revealed progressive loss of productive reactive geometry in many complexes. These observations motivated the comparative analyses of benchmark sotorasib systems, geometry-guided

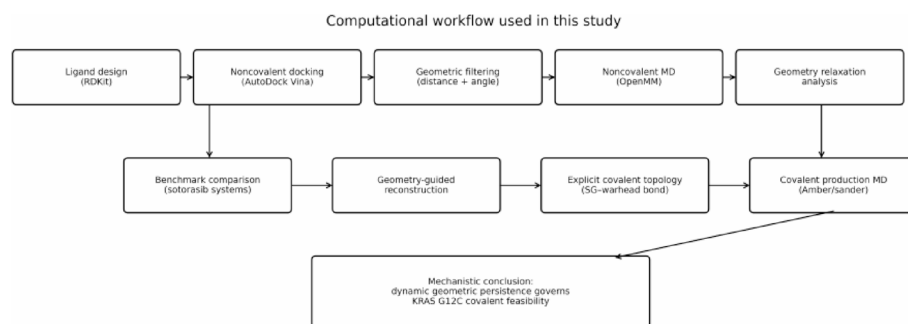


Fig. 1 Workflow integrating docking, geometric filtering, noncovalent MD, benchmark guided reconstruction, and covalent production MD

reconstruction strategies, and explicitly covalent production simulations presented in the following sections.

Figure 1 illustrates computational workflow integrating ligand design, docking, geometric filtering, non-covalent molecular dynamics, benchmark-guided reconstruction, and covalent production simulations to evaluate covalent feasibility in KRAS G12C systems. Initial docking and geometric filtering identified ligand poses with apparent near-reactive sulfur–electrophile proximity, whereas subsequent non-covalent molecular dynamics simulations revealed relaxation of productive reactive geometry. Comparative analysis of benchmark sotorasib systems and geometry-guided covalent reconstruction was subsequently used to evaluate whether explicit covalent stabilization could preserve dynamically persistent reactive alignment.

3.1 Iterative scaffold optimization reveals persistent geometric constraints in the KRAS G12C pocket

The KRAS G12C mutant contains a transiently accessible Switch-II pocket adjacent to the reactive CYS12 residue, enabling selective covalent inhibition through electrophile-directed targeting [3, 4]. Clinically validated inhibitors such as sotorasib exploit this pocket by combining non-covalent stabilization with precise electrophile preorganization toward the cysteine thiol [4, 5]. Because productive covalent engagement depends strongly on reactive geometry rather than binding affinity alone, the present study systematically investigated how progressive scaffold modifications influence electrophile positioning within the KRAS G12C binding environment. Ligands were developed through iterative design rounds denoted by the prefix “R”, corresponding to successive generations of scaffold refinement, while individual analogues within each round were identified using the suffix “A”. Early design cycles (R1–R3) focused on establishing baseline occupancy of the Switch-II pocket using acrylamide-containing scaffolds with systematic variation in aromatic substitution, linker composition, and electrophile topology. Although several ligands achieved moderate docking affinity, electrophile positioning remained consistently unfavorable relative to the sulfur atom of CYS12. Linker extension and scaffold reorientation improved pocket accommodation but failed to direct the acrylamide warhead into a productive pre-reactive orientation.

The experimentally validated binding geometry of sotorasib was used as a structural reference throughout the docking and geometry-guided analyses to evaluate electrophile positioning relative to Cys12. Comparative analysis showed that several custom ligands could physically access the cysteine-adjacent region of the Switch-II pocket, indicating

that the principal limitation was not steric reach alone, but rather electrophile orientation and geometric preorganization. Guided by these observations, the R4 series introduced redesigned linker topologies intended to redirect the acrylamide warhead toward the reactive region while preserving favorable Switch-II pocket interactions. This scaffold reorganization produced the first meaningful improvement in cysteine accessibility and established a structural framework for subsequent optimization. Building upon this architecture, the R5 series explored aromatic substitution patterns surrounding a fluoro-phenyl anchoring motif. Among these ligands, R5_A3 emerged as an important intermediate scaffold, combining stable pocket occupancy with one of the closest electrophile approaches observed at that stage of the campaign. Nevertheless, the ligand still maintained a non-reactive SG–electrophile separation of approximately 6.3 Å and formed stabilizing interactions near GLY60, suggesting trapping in a non-productive binding orientation.

The mechanistic significance of this limitation became clearer in the R6 series, where redesign efforts attempted to reduce non-productive linker interactions and improve electrophile presentation toward CYS12 while preserving the favorable fluoro-phenyl anchor. Although compounds such as R6_A0 retained favorable docking affinity, they failed to achieve productive reactive geometry, demonstrating that binding affinity and covalent alignment are governed by distinct structural determinants. These observations established that ligand positioning within the KRAS G12C pocket is dominated by stabilizing non-covalent interactions that can compete directly with productive electrophile orientation. Subsequent scaffold generations therefore shifted toward explicit geometric control of electrophile presentation. The R11_R13 series investigated linker extension, conformational flexibility, amide inversion, and steric steering strategies to improve directional alignment toward the reactive cysteine. Although these modifications improved access to the reactive region, increased linker flexibility frequently destabilized electrophile orientation and favored non-productive conformations.

Further refinement in the R14 and R15 series introduced cyano-acrylamide warheads and substituent-driven preorganization strategies designed to improve electrophile reactivity and directional bias. These modifications progressively reduced electrophile–cysteine separation and improved near-reactive alignment, indicating partial enhancement of pre-reactive organization within the Switch-II pocket. Among the later-stage candidates, R16_A4 represented one of the first ligands to reproducibly achieve near-reactive static docking geometry while maintaining favorable pocket occupancy. Continued linker and core reorganization in the R17 series improved electrophile proximity further; however, simultaneous optimization of distance and Bürgi–Dunitz-consistent attack geometry remained unresolved, with several ligands exhibiting persistent angular misalignment despite improved cysteine accessibility.

The most advanced scaffold emerged in the R18 series, particularly R18_A4, which combined a cyano-acrylamide electrophile with optimized meta-cyano and para-fluoro substitution patterns. This ligand consistently achieved one of the closest near-reactive docking geometries observed throughout the study while maintaining favorable occupancy of the Switch-II pocket. Nevertheless, none of the evaluated docking poses fully reproduced the productive covalent geometry observed in crystallographic KRAS G12C inhibitor complexes [4]. Collectively, these results demonstrate that iterative scaffold optimization substantially improved electrophile proximity and directional organization

within the KRAS G12C binding pocket. However, the progressive emergence of near-reactive docking poses also revealed that static docking alone is insufficient to determine whether these geometries remain dynamically stable or chemically productive under explicit-solvent conditions. These observations motivated subsequent molecular dynamics analyses aimed at evaluating the persistence of near-reactive configurations over time (Tables 1 and 2).

Covalent feasibility in the KRAS G12C system appears to depend on geometric pre-organization and dynamic persistence of reactive alignment rather than binding affinity alone. Across successive optimization rounds, ligand modifications progressively improved electrophile proximity and partially enhanced directional alignment toward CYS12; however, near-reactive docking geometries were frequently destabilized during molecular dynamics simulations. These observations indicate that productive covalent engagement requires simultaneous satisfaction of distance, trajectory, and conformational stability within the constrained Switch-II pocket environment. In contrast, the same scaffold maintained more stable pre-reactive positioning in BTK, supporting the concept that covalent feasibility is strongly target-dependent and governed by the interplay between ligand architecture and local binding-site dynamics [12–20]. Collectively,

Table 1 Representative scaffold evolution and geometric outcomes across the KRAS G12C ligand design trajectory

Generation	Representative ligand	Key design modification	Representative observation	Mechanistic interpretation
R5	R5_A3	Fluorophenyl acrylamide scaffold	Closest SG–electrophile distance ~6.26 Å with strong GLY60 interaction	Pocket occupancy achieved, but warhead remained geometrically displaced from CYS12
R11	R11_A0	Extended linker acrylamide	Improved reach toward reactive region but unstable orientation	Increased flexibility did not preserve productive geometry
R12	R12_A2	Ortho-methyl steric steering	Partial electrophile reorientation toward CYS12	Steric effects modestly influenced attack trajectory
R13	R13_A0	Flexible linker redesign	Near-reactive geometries transient and unstable	Excess linker mobility favored non-productive conformations
R14	R14_A1	Cyano-acrylamide warhead introduction	Reduced electrophile–cysteine separation	Increased electrophilicity alone insufficient for productive alignment
R15	R15_A1	α-Methyl and ortho-fluoro substitution	Improved directional bias toward CYS12	Scaffold preorganization partially enhanced
R16	R16_A4	meta-CN/meta-F cyanoacrylamide	Reproducible near-reactive docking geometry	Significant improvement in static electrophile positioning
R17	R17_A13	aminobenzyl kink variant	Improved proximity with persistent angular misalignment	Simultaneous optimization of distance and trajectory remained unresolved
R18	R18_A4	meta-CN/para-F cyanoacrylamide	Strong pocket occupancy with closest near-reactive geometry	Selected for dynamic evaluation by molecular dynamics
Benchmark	Sotorasib	Clinically validated KRAS G12C inhibitor	Productive crystallographic covalent geometry	Reference framework for electrophile preorganization
Final	Manually reconstructed ligand	Exit-vector-guided covalent reconstruction	Preserved near-reactive covalent alignment	Geometry-guided reconstruction improved reactive organization

Table 2 Quantitative geometric evaluation of representative KRAS G12C ligand scaffolds across successive optimization rounds

Representative ligands from different rounds	Affinity (kcalmol ⁻¹)	SG–Electrophile Distance (Å)	Attack Geometry	Reactive classification
R5_A3	-7.26	6.26	Poor	Non-reactive
R11_A0	~ -6.1	8–11	Poor	Non-reactive
R12_A2	~ -6.5	5–8	Suboptimal	Near-reactive
R14_A1	~ -7.0	3.5–4.0	Poor	Near-reactive
R15_A1	~ -7.2	3.5–4.5	Suboptimal	Near-reactive
R16_A4	-7.95	3.69	Poor	Near-reactive
R17_A13	-8.23	3.84	Poor	Near-reactive
R18_A4	~ -8.0	1.65	Near optimal	Near-reactive
R18_reactive pose	-11.90	1.35	Suboptimal	Non-reactive
Benchmark sotorasib	Literature	~ 1.8	Productive	Reactive

Binding affinity, electrophile–CYS12 sulfur separation, and qualitative attack geometry were evaluated to assess covalent feasibility during iterative scaffold refinement. Progressive ligand optimization improved electrophile proximity and partially enhanced directional alignment toward CYS12; however, productive covalent geometry was not consistently achieved across non-covalent docking ensembles. Although late-stage ligands such as R18_A4 reached near-reactive sulfur distances, persistent geometric misalignment remained a dominant limitation relative to the productive covalent geometry observed in the benchmark inhibitor sotorasib

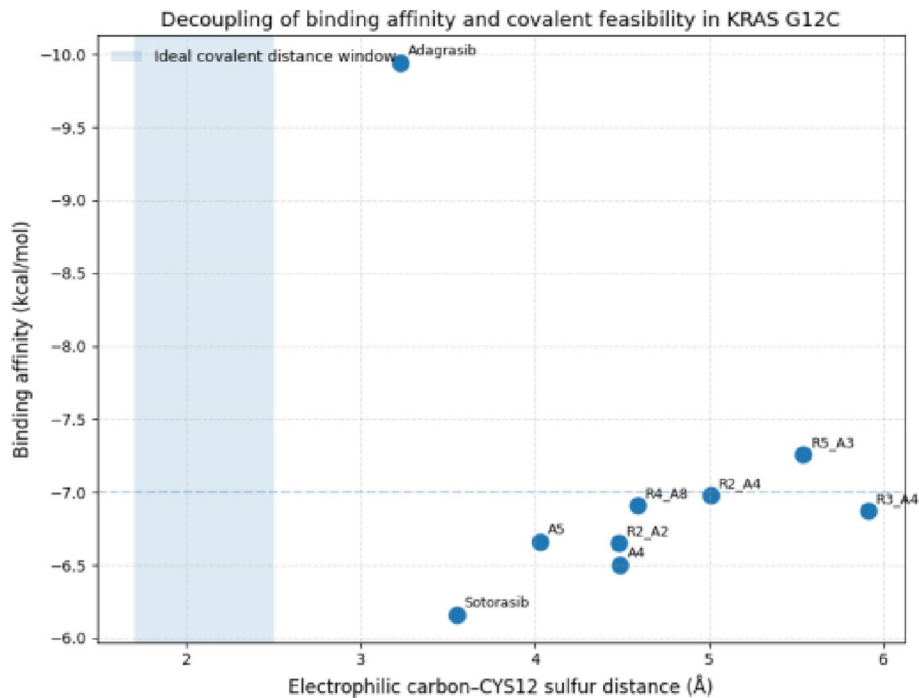


Fig. 2 Relationship between binding affinity and electrophilic carbon–CYS12 sulfur distance for representative KRAS G12C ligands. The shaded region denotes the ideal covalent distance window. While several ligands exhibit favorable binding affinities, none satisfy the geometric requirements for covalent bond formation. This highlights a clear decoupling between affinity and covalent feasibility

these findings suggest that successful KRAS G12C covalent inhibitors require rigid electrophile preorganization capable of preserving productive attack geometry throughout dynamic conformational sampling, highlighting limitations of affinity-driven docking approaches that do not explicitly account for reactive-state stability [3, 4, 10, 11] (Fig. 2)

Despite progressive improvements in electrophile proximity across successive optimization rounds, productive covalent geometry remained difficult to achieve within the

KRAS G12C Switch-II pocket. Several late-stage ligands, including R16_A4, R17_A13, and R18_A4, approached near-reactive sulfur separation distances while maintaining favorable pocket occupancy; however, electrophile orientation frequently remained misaligned relative to the preferred nucleophilic attack trajectory. In many cases, ligand conformations favored energetically stable non-covalent interactions that redirected the electrophilic warhead away from productive alignment with CYS12. Refinement of the docking protocol and covalent docking constraints improved local sampling near the reactive residue but did not fully resolve these geometric limitations. Collectively, these observations indicate that covalent feasibility in KRAS G12C is governed not only by electrophile proximity, but also by the ability of the ligand scaffold to preserve productive reactive alignment within the conformationally constrained Switch-II binding environment.

3.2 Molecular dynamics reveals limited stability of near-reactive states

To evaluate whether docking-derived geometries remain dynamically stable, molecular dynamics simulations were performed on selected KRAS G12C–ligand complexes [15, 21]. Although several ligands initially occupied near-reactive configurations, these states were not preserved during simulation. Electrophile–cysteine distances increased progressively over time, with final separations typically reaching 5–6 Å and average distances substantially exceeding the reactive regime (Fig. 3). This loss of reactive proximity occurred despite continued ligand occupancy within the Switch-II pocket, indicating that favorable docking poses do not necessarily correspond to stable pre-reactive conformations. In contrast, crystallographic KRAS G12C inhibitors maintain

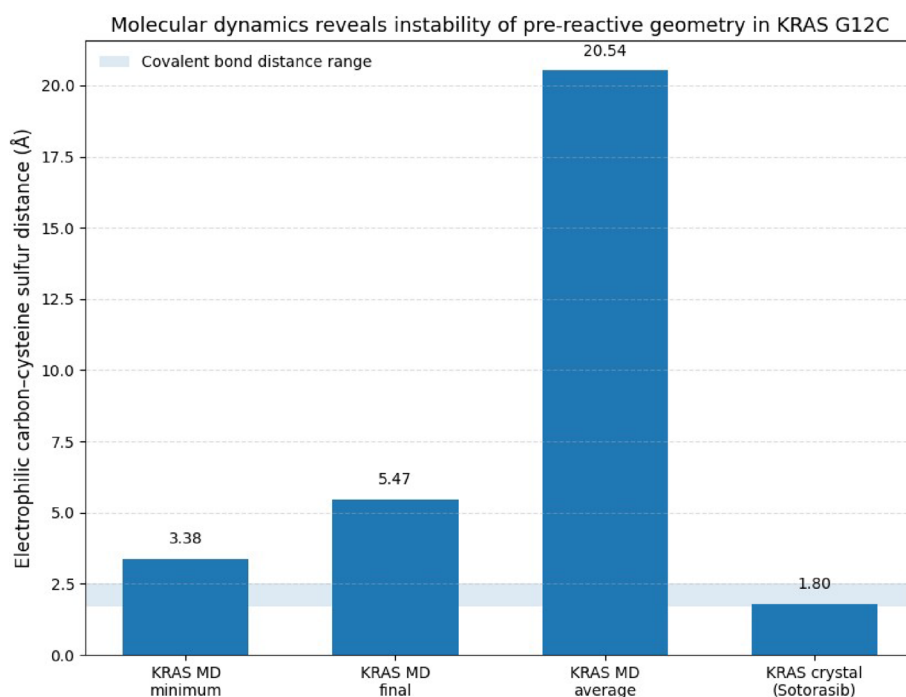


Fig. 3 Molecular dynamics analysis of electrophile–cysteine distance in the KRAS G12C system. Although the ligand transiently approaches near-reactive distances, this configuration is not maintained, with increasing separation over time and a large average distance. In contrast, the experimentally observed covalent complex exhibits a stable bond-length distance within the ideal covalent range. These results demonstrate that docking-derived geometries do not correspond to dynamically stable pre-reactive states

electrophile–thiol separations consistent with productive covalent geometry [4]. These observations suggest that covalent feasibility in the KRAS G12C system depends not only on transient geometric accessibility, but also on the dynamic persistence of reactive alignment under explicit-solvent conditions.

Analysis of CYS12 sulfur mobility further showed that local fluctuations of the SG atom remained spatially restricted throughout the trajectory, with displacements confined to approximately 2 Å (Fig. 4). These limited motions were insufficient to compensate for electrophile misalignment or restore productive attack geometry once reactive configurations relaxed during simulation. Together, the MD results indicate that near-reactive docking poses sampled in the KRAS G12C pocket correspond primarily to transient rather than dynamically stable covalent precursor states.

Together, docking, geometric analysis, and molecular dynamics consistently indicate that covalent feasibility in KRAS G12C is limited by intrinsic geometric and dynamic constraints that are not resolved through ligand optimization. These observations suggest that successful KRAS G12C inhibition requires precise geometric preorganization rather than proximity alone, emphasizing the importance of incorporating geometric and preorganization metrics into covalent inhibitor design.

3.3 Dynamic instability of near-reactive geometry in the R18_A4 complex

R18_A4 was selected for molecular dynamics evaluation following docking analyses that identified one of the closest near-reactive geometries observed during iterative scaffold optimization. Although the ligand remained stably associated with the KRAS G12C Switch-II pocket throughout the simulation, trajectory analysis revealed progressive loss of productive reactive geometry. The SG–electrophile distance fluctuated predominantly within the 6–8 Å range and did not persist within the reactive regime required for covalent bond formation. Simultaneously, the corresponding nucleophilic attack trajectory

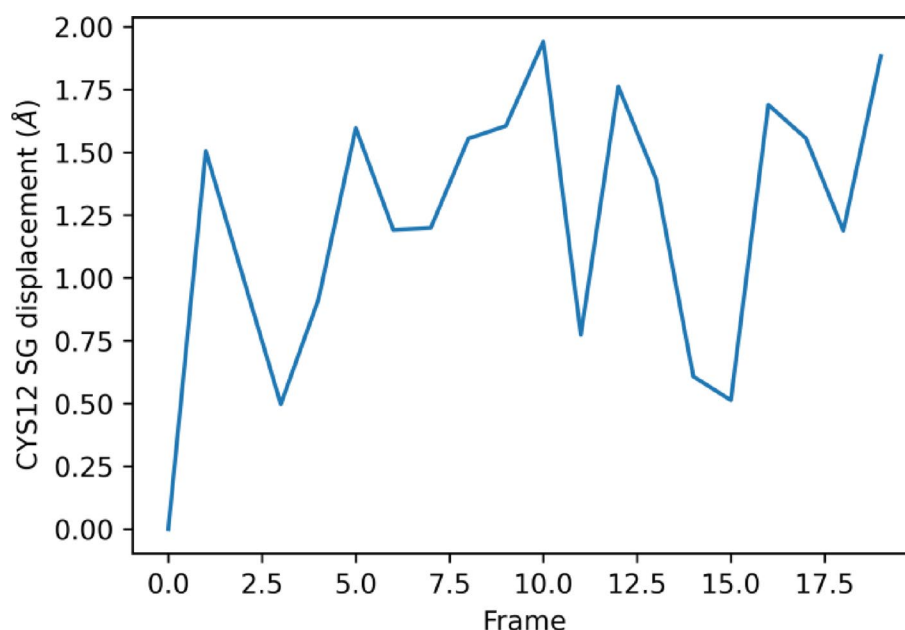


Fig. 4 Displacement of the CYS12 sulfur atom over the aligned KRAS G12C molecular dynamics trajectory. The SG atom exhibits limited positional fluctuations (≤ 2 Å), indicating restricted local mobility. Such constrained dynamics are insufficient to reorient electrophile trajectories toward a productive Bürgi–Dunitz attack geometry, supporting the persistence of geometric limitations in the KRAS G12C binding environment

sampled broad angular distributions, with only transient occupancy of geometrically favorable orientations.

Importantly, favorable angular alignment and reactive sulfur proximity were not maintained simultaneously during the trajectory. Thus, despite achieving near-reactive docking configurations and stable pocket occupancy, R18_A4 failed to preserve the coupled geometric requirements necessary for productive covalent engagement. These observations indicate that apparent docking compatibility in KRAS G12C systems does not necessarily translate into dynamically stable pre-reactive organization. Instead, covalent feasibility appears to depend critically on the persistent maintenance of both electrophile proximity and reactive attack geometry throughout molecular motion.

3.4 Structure-guided reconstruction and dynamic evaluation of a pre-reactive KRAS G12C ligand

Following repeated failure of docking-derived ligands to reproduce a covalent-competent orientation within the KRAS G12C pocket, a structure-guided reconstruction strategy was adopted to evaluate whether a near-reactive geometry could be generated directly from the crystallographic binding framework of sotorasib [4, 5]. Earlier scaffold generations achieved stable occupancy of the Switch-II region but consistently failed to position the electrophilic warhead along a productive trajectory toward Cys12, indicating that the principal limitation was not binding affinity alone, but incorrect electrophilic exit-vector orientation [1, 6]. Consequently, the design problem was reformulated from affinity optimization toward preservation of a productive electrophilic reaction vector.

To address this limitation, the crystallographic KRAS G12C–sotorasib complex was used as a geometric reference. The covalent attachment atom of sotorasib (C25) was identified as the electrophilic exit point, and the corresponding Cys12 SG→C25 reaction vector was calculated as [0.5428, 0.6048, − 0.5827]. New atoms were then manually reconstructed along this trajectory to approximate a near-attack configuration while preserving the original sotorasib-derived binding framework. In the resulting engineered geometry, the reconstructed electrophilic carbon (X26/C31) was positioned approximately 2.40 Å from the sulfur atom of Cys12, while the corresponding SG–C31–C25 attack angle was maintained near the Bürgi–Dunitz region ($\sim 105^\circ$), consistent with geometries associated with productive nucleophilic attack in covalent complexes [4, 10, 11].

One representative manually reconstructed ligand was subsequently selected for dynamic evaluation using explicit-solvent non-covalent molecular dynamics simulations. The tested scaffold was a sotorasib-derived acrylamide analogue represented by the SMILES string:

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C = CC(=O)N[C@H]1CN(C(=O)CC)CCN1c1nc(=O)n(c2c(C)ccnc2C(C)C)c2nc(cc12)F)c1c(F)cccc1O.
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This manually engineered ligand was not intended to represent a fully optimized therapeutic candidate, but rather to test whether restoration of the crystallographic covalent trajectory alone would be sufficient to preserve reactive geometry during molecular dynamics simulations.

Equilibration and production dynamics progressively displaced the electrophilic warhead away from Cys12, with sulfur–carbon distances predominantly sampling a non-reactive range of approximately 8–10 Å throughout the production trajectory. Notably,

the predefined reactive distance threshold (<4 Å) was not sampled during production dynamics despite the manually imposed starting geometry. Although the ligand remained associated with the KRAS binding region, sustained pre-reactive geometry was not preserved under equilibrium conditions. These observations indicate that restoration of a geometrically plausible covalent trajectory alone does not guarantee persistence of a covalent-ready state under equilibrium conditions.

These findings demonstrate that restoration of a crystallographically correct electrophilic trajectory is geometrically achievable, but dynamically unstable in the absence of sufficient energetic stabilization by the ligand scaffold. Collectively, the results indicate that covalent feasibility in KRAS G12C systems depends not only on access to a near-reactive geometry, but also on the thermodynamic capacity of the molecular scaffold to maintain that configuration under dynamic conditions.

3.5 Benchmark molecular dynamics analysis of sotorasib

3.5.1 Non-covalent molecular dynamics of sotorasib reveals dynamic instability of the reactive configuration

To establish a mechanistic benchmark for reactive geometry preservation in KRAS G12C systems, molecular dynamics simulations were performed on a reconstructed non-covalent KRAS–sotorasib complex in which the crystallographic covalent linkage between Cys12 and the ligand warhead was removed. Sotorasib was selected as the reference system because it represents a clinically validated KRAS G12C covalent inhibitor with an experimentally resolved productive binding geometry [4, 5]. In the crystallographic complex, the sulfur atom of Cys12 forms a covalent bond with the electrophilic carbon atom C25 of sotorasib at an approximate sulfur–carbon distance of 1.81 Å. The corresponding SG→C25 reaction vector extracted from the crystal structure was calculated as [0.5428, 0.6048, −0.5827], representing the experimentally observed trajectory associated with productive covalent engagement (Figs. 5, 6, 7 and 8).

Following removal of the covalent bond, the reconstructed non-covalent complex was subjected to explicit-solvent molecular dynamics simulations under equilibrium conditions. Reactive geometry was evaluated using two descriptors: (i) the distance between the sulfur atom of Cys12 (SG) and the electrophilic carbon of sotorasib (C25), and (ii) the SG–C25–C24 attack angle associated with nucleophilic approach during Michael addition. Throughout the trajectory, the SG–C25 distance fluctuated approximately between 3.5 and 5.8 Å, indicating transient sampling of near-reactive proximity but failure to maintain distances consistent with stable covalent engagement. Similarly, the SG–C25–C24 attack angle sampled a broad range of approximately 75°–128°, with only transient occupancy of Bürgi–Dunitz-like orientations expected for productive nucleophilic attack [10, 11]. Importantly, favorable angular conformations were not accompanied by sustained short sulfur–carbon distances, demonstrating loss of coupled reactive geometry during unconstrained non-covalent dynamics.

As the simulation progressed, the ligand relaxed away from the crystallographic covalent configuration despite remaining associated with the switch-II binding region. These observations suggest that although sotorasib achieves productive covalent engagement experimentally, the crystallographic reactive configuration is not stably maintained under equilibrium non-covalent conditions after removal of the covalent linkage. The results therefore indicate that covalent stabilization itself plays a critical role in

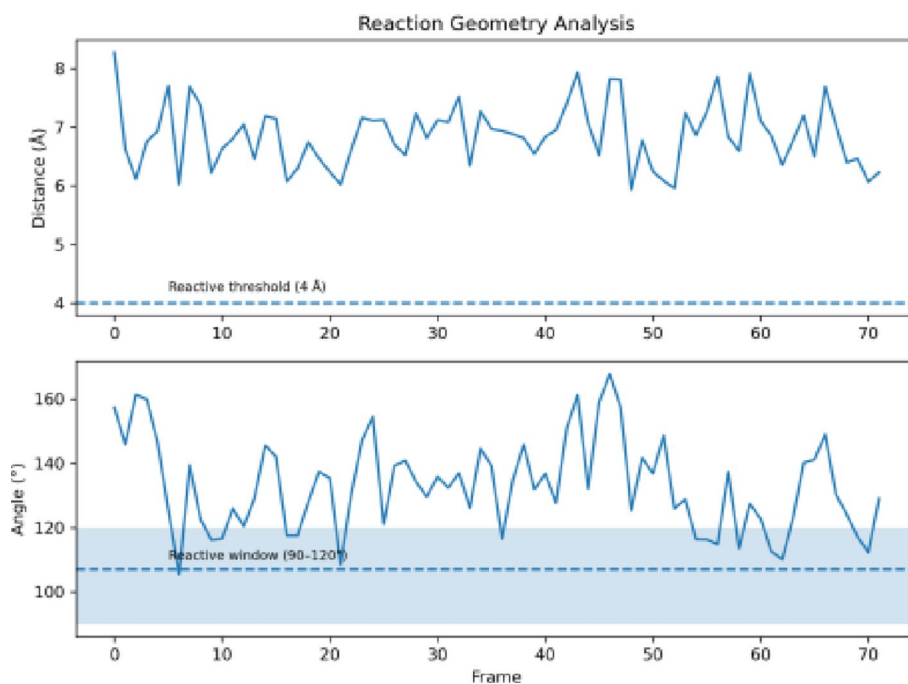


Fig. 5 Evolution of the SG-C31 distance (upper panel) and SG-C31-C25 attack angle (lower panel) during non-covalent molecular dynamics simulations of the R18_A4 KRAS G12C complex. The dashed line indicates the reactive distance threshold (4 Å), and the shaded region denotes the favorable Bürgi-Dunitz attack-angle window (90–120°). Although transiently favorable attack angles were occasionally sampled, productive distance and angle were not maintained simultaneously, demonstrating that productive reactive geometry is not persistently maintained during non-covalent molecular dynamics

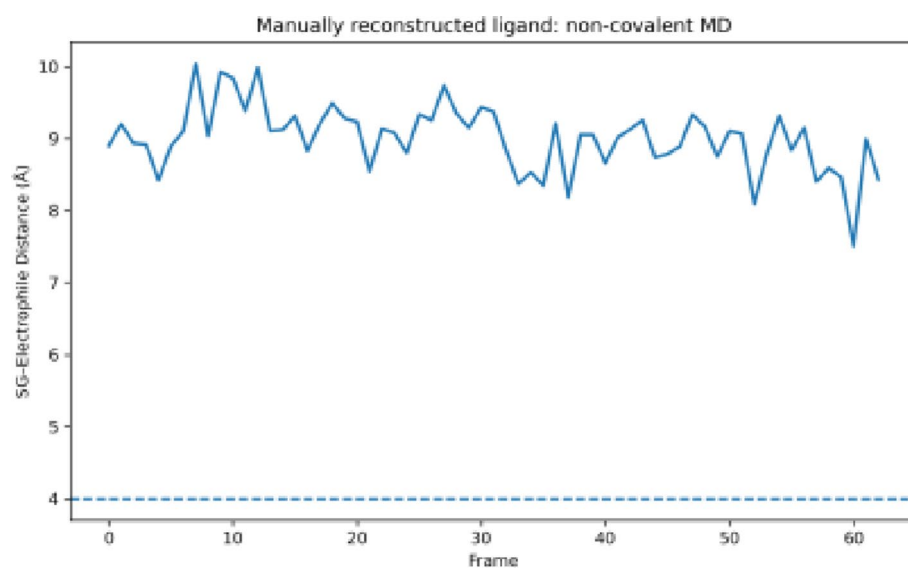


Fig. 6 Time evolution of the Cys12 SG-electrophile distance during non-covalent molecular dynamics simulations of the manually reconstructed Sotorasib-derived ligand. Although the initial configuration was engineered from the crystallographic Sotorasib exit vector, the sulfur-electrophile separation progressively relaxed toward a non-reactive equilibrium range (~8–10 Å) and failed to sample the predefined reactive threshold (<4 Å) during production dynamics. These results indicate that restoration of a geometrically plausible covalent trajectory alone is insufficient to stabilize a reactive configuration under non-covalent conditions

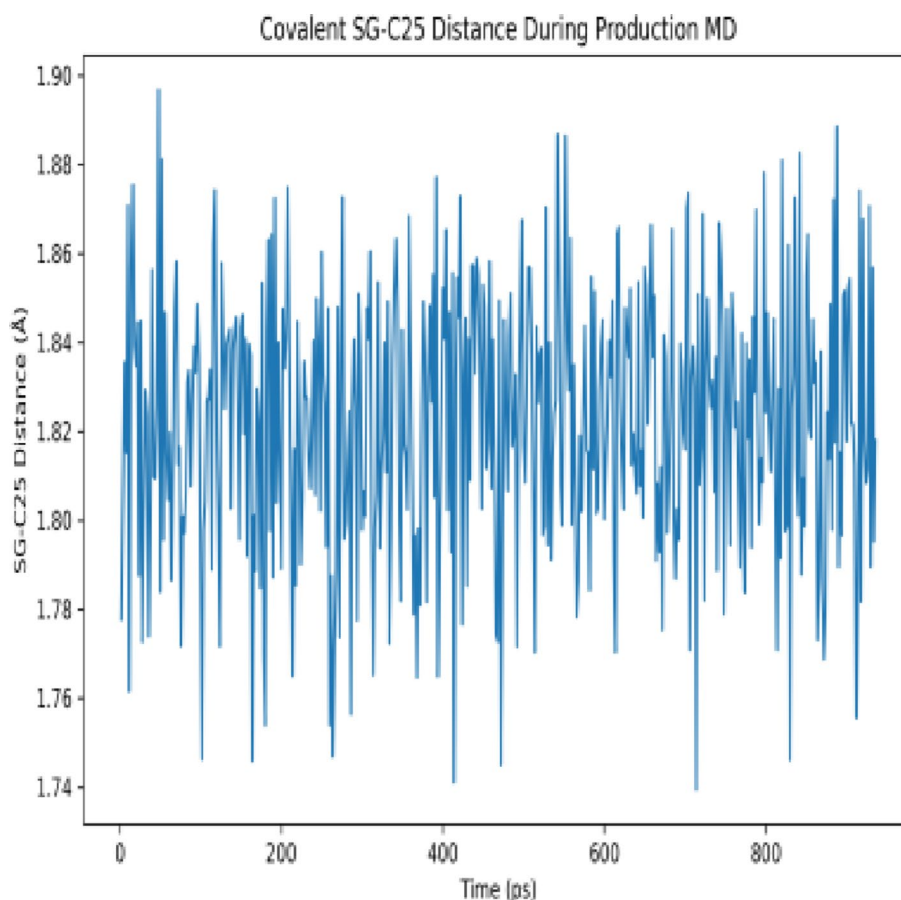


Fig. 7 Time evolution of the covalent SG–C25 bond distance during production molecular dynamics simulations of the KRAS–sotorasib complex. The sulfur–carbon distance remained tightly preserved near the expected covalent bond length throughout approximately 830 ps of production dynamics, fluctuating only within a narrow range of ~1.74–1.90 Å without evidence of bond elongation or dissociation. These results demonstrate persistent stabilization of the covalent linkage under equilibrium conditions

preserving reactive alignment within the KRAS G12C binding pocket. This benchmark analysis provides an important mechanistic reference for the present study by demonstrating that ligand localization within the switch-II pocket alone is insufficient to ensure persistent covalent-ready geometry. Rather, productive covalent feasibility appears to depend on dynamic stabilization of both reactive distance and electrophilic attack orientation simultaneously.

3.5.2 Covalent molecular dynamics simulations demonstrate stable preservation of reactive geometry in the *kras*–sotorasib complex

To determine whether covalent attachment stabilizes reactive geometry within the KRAS G12C switch-II pocket, extended production molecular dynamics simulations were performed on the experimentally validated covalent KRAS–sotorasib complex. In contrast to the reconstructed non-covalent benchmark system described above, the covalent linkage between the sulfur atom of Cys12 (SG) and the electrophilic carbon atom of sotorasib (C25) was explicitly maintained throughout the simulation. The resulting covalent complex was subjected to sequential production molecular dynamics

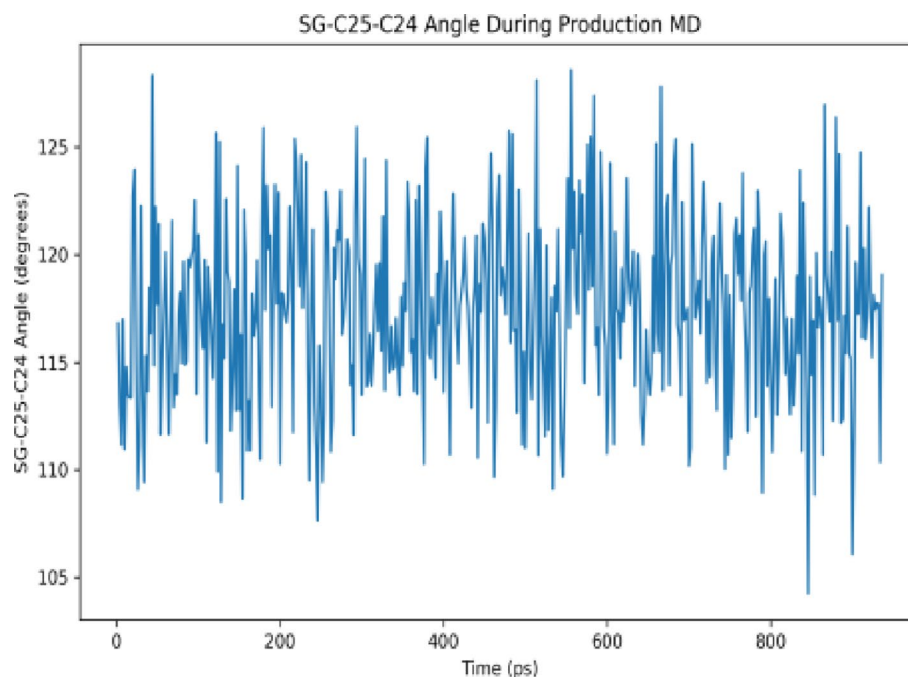


Fig. 8 Time evolution of the SG–C25–C24 attack angle during production molecular dynamics simulations of the covalent KRAS–sotorasib complex. The covalent attack geometry remained consistently preserved throughout the trajectory, with angular fluctuations centered near the productive Bürgi–Dunitz region ($\sim 117^\circ$). The absence of progressive angular distortion indicates stable maintenance of reactive alignment within the KRAS G12C switch-II pocket

simulations under explicit-solvent conditions, yielding approximately 830 ps of cumulative production sampling across 418 analyzed trajectory frames.

Throughout the production trajectory, the covalent KRAS–sotorasib complex exhibited remarkable geometric and structural stability. The SG–C25 covalent bond distance remained tightly centered around the expected sulfur–carbon bond length, with an average distance of 1.82 Å and a narrow fluctuation range of 1.74–1.90 Å. Importantly, no analyzed frame exceeded a sulfur–carbon separation of 2.0 Å, indicating complete preservation of the covalent linkage during the simulation. Similarly, the SG–C25–C24 attack geometry remained highly stable, with an average angle of 117.48° and a thermally reasonable fluctuation range of $107.7\text{--}128.6^\circ$. No evidence of progressive bond elongation, geometric drift, or destabilization of the covalent configuration was observed during late-stage production dynamics.

Global structural analysis further confirmed stability of the KRAS protein throughout the trajectory. Backbone RMSD values remained low, averaging approximately 0.77 Å with a maximum value of only 1.02 Å, indicating preservation of the overall protein fold during production dynamics. The radius of gyration also remained highly stable, averaging 15.30 Å with minimal fluctuation, consistent with maintenance of a compact and structurally equilibrated protein state. Per-residue RMSF analysis showed moderate local flexibility primarily within terminal regions, without evidence of abnormal residue-level instability.

Hydrogen-bond analysis revealed only transient ligand–protein hydrogen bonding interactions with low occupancies ($\sim 0.24\%$), suggesting that persistent classical hydrogen bonds are not the dominant stabilizing factor within the complex. Instead, the

observed long-timescale stability appears to arise primarily from covalent anchoring of the ligand within the switch-II pocket, supplemented by favorable hydrophobic and van der Waals interactions.

These results sharply contrast with the behavior observed in the non-covalent and manually reconstructed systems, both of which progressively relaxed away from pre-reactive conformations during equilibration and production dynamics. Collectively, the present simulations demonstrate that experimentally validated KRAS G12C covalent inhibitors maintain reactive geometry not through transient electrophile proximity alone, but through sustained energetic stabilization of the covalent state itself. The covalent KRAS–sotorasib complex therefore serves as a mechanistic reference demonstrating that persistent preservation of reactive geometry within the KRAS G12C binding pocket is strongly coupled to covalent stabilization (Table 3).

SEM denotes the standard error of the mean calculated from the analyzed trajectory frames (or residues for RMSF). Analysis scope indicates the number of trajectory frames used for each individual metric.

3.6 Covalent stabilization of a manually reconstructed KRAS G12C ligand geometry

Following the non-covalent molecular dynamics analyses described above, an explicit covalent reconstruction strategy was implemented to determine whether covalent anchoring could stabilize a manually engineered reactive geometry that previously failed under equilibrium non-covalent conditions. The manually reconstructed ligand was derived from the Sotorasib scaffold through structure-guided modification of the covalent attachment region while preserving the experimentally validated switch-II pocket binding framework. In prior non-covalent simulations, this reconstructed system rapidly relaxed away from the engineered pre-reactive configuration, indicating that geometric accessibility alone was insufficient to preserve reactive alignment dynamically. To evaluate whether covalent stabilization could rescue this previously unstable configuration, an explicit sulfur–carbon covalent linkage was introduced between the sulfur atom of KRAS G12C Cys12 and the reconstructed electrophilic carbon atom (C31) prior to molecular dynamics simulations. The benchmark covalent Sotorasib system and the manually reconstructed covalent ligand analyzed in the present study are summarized in Table 4.

To facilitate structural comparison between the experimentally validated benchmark and the structure-guided reconstructed system, the corresponding two-dimensional chemical structures are presented in Figs. 9 and 10. These structures were generated directly from the SMILES representations listed in Table 4 using RDKit.

Following topology generation and explicit covalent parameterization, the manually reconstructed KRAS G12C covalent complex was subjected to a full molecular dynamics workflow consisting of energy minimization, gradual heating, equilibration, and ~2 ns

Table 3 Summary of molecular dynamics stability metrics for the covalent KRAS–sotorasib complex

Parameter	Average $\pm$ SEM	Minimum	Maximum	Analysis Scope
SG–C25 Distance (Å)	1.8219 $\pm$ 0.0014	1.7394	1.8967	418 frames
SG–C25–C24 Angle (°)	117.48 $\pm$ 0.20	107.66	128.57	418 frames
Backbone RMSD (Å)	0.773 $\pm$ 0.0050	0.000	1.019	418 frames
Radius of Gyration (Å)	15.295 $\pm$ 0.0022	15.181	15.419	418 frames
Ca RMSF (Å)	4.309 $\pm$ 0.0420	3.120	5.357	167 residues

Table 4 Covalent KRAS G12C ligand systems analyzed in the present study.

System description	Scientific role	Smiles
Benchmark Covalent sotorasib	Experimentally validated reference covalent KRAS G12C inhibitor used as the mechanistic benchmark for stable reactive geometry preservation.	<chem>C[C@H]1CN(C(=O)CC)CCN1c1nc(=O)n(c2c(C)ccnc2C(C)C)c2nc(c(cc12)F)c1c(F)cccc1O</chem>
Manually Reconstructed Covalent Ligand	Structure-guided reconstructed scaffold used to evaluate whether explicit covalent anchoring can stabilize a previously unstable pre-reactive geometry under molecular dynamics conditions.	<chem>C[C@H]1 C[N@@]([C@H]2O[C@@]3([C@H]4[C@@H]2C4)ON3)CCN1c1nc(=O)n(c2c(C)ccnc2C(C)C)c2nc(c(cc12)F)c1c(F)cccc1O</chem>

The benchmark covalent Sotorasib scaffold represents the experimentally validated reference inhibitor, whereas the manually reconstructed ligand was engineered as a mechanistic test system to evaluate whether covalent anchoring can stabilize a previously unstable reactive geometry

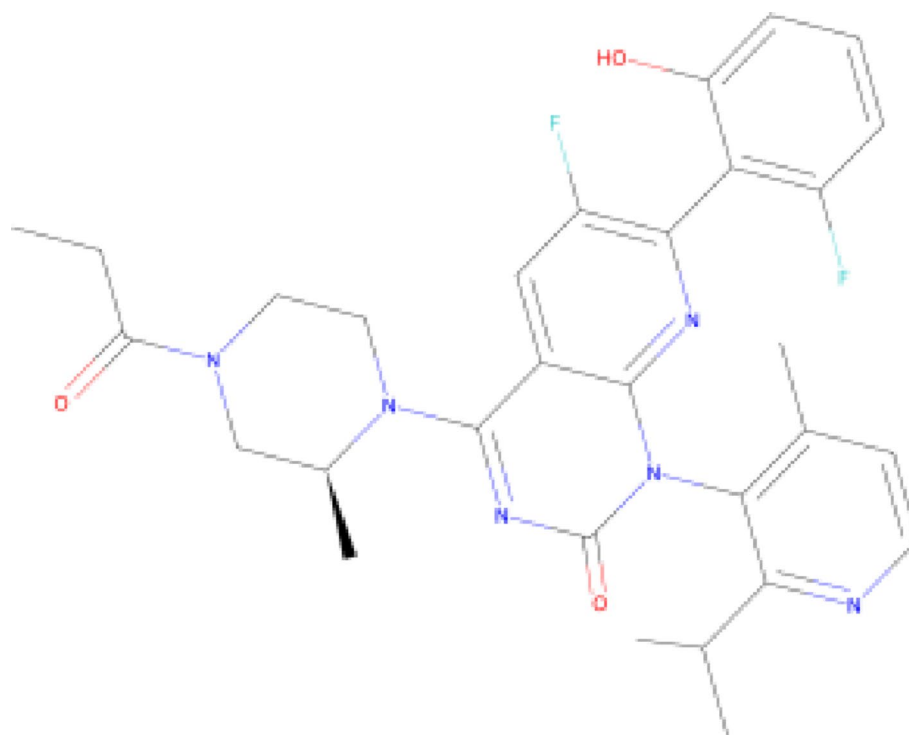


Fig. 9 Two-dimensional chemical structure of sotorasib, the experimentally validated KRAS G12C covalent inhibitor used as the benchmark reference throughout this study for docking validation, geometry-guided reconstruction, and covalent molecular dynamics simulations

of production molecular dynamics under explicit-solvent NPT conditions. Throughout the production simulation, the system remained thermodynamically stable, maintaining temperatures near the target value of 300 K and solvent densities consistently close to 1.0 g cm^{-3} . No evidence of structural instability, ligand dissociation, simulation drift, or periodic box collapse was observed during the trajectory, indicating that the reconstructed covalent complex remained well equilibrated over the nanosecond timescale.

To directly evaluate preservation of the engineered covalent geometry, the sulfur-carbon linkage between KRAS G12C Cys12 (SG) and the reconstructed electrophilic carbon atom (C31) was monitored throughout the complete production trajectory (~ 2 ns; 1664 analyzed frames). The SG–C31 bond distance remained highly stable, with a

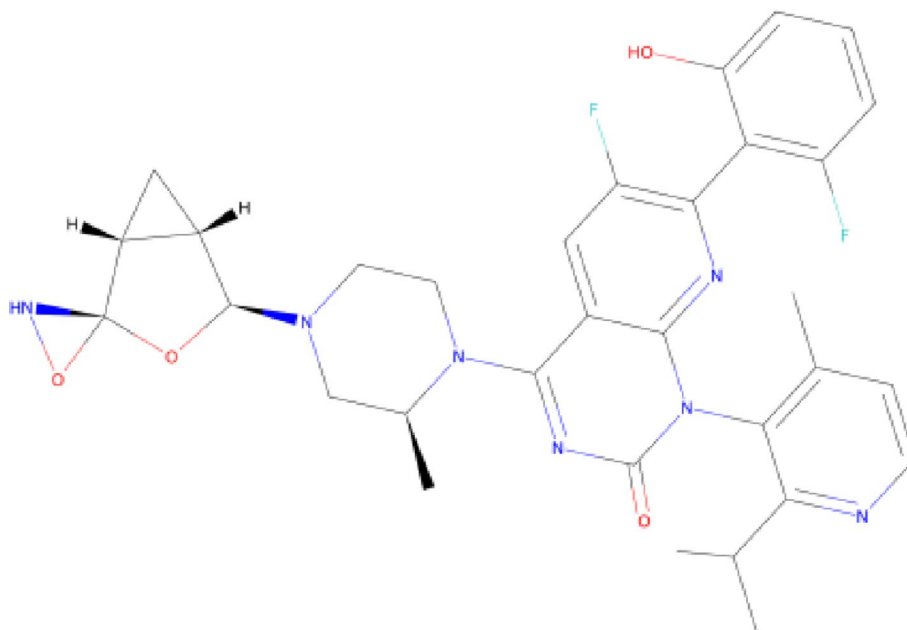


Fig. 10 Two-dimensional chemical structure of the geometry-guided manually reconstructed ligand used for mechanistic evaluation of covalent feasibility in KRAS G12C. The ligand was reconstructed based on the benchmark sotorasib binding geometry to preserve chemically favorable sulfur–electrophile alignment and served as the starting structure for covalent molecular dynamics simulations

mean value of 1.844 ± 0.032 Å ($SE = 0.00078$ Å) and an observed range of 1.748–1.950 Å. These values are consistent with a persistent covalent sulfur–carbon bond and showed no evidence of progressive bond elongation or destabilization during the simulation. In parallel, the SG–C31–C32 bond angle remained centered at $119.56 \pm 3.12^\circ$ ($SE = 0.077^\circ$), fluctuating within a narrow range of 109.77 – 129.20° throughout the trajectory. The limited variability observed for both geometric descriptors indicates that the engineered reactive arrangement remained preserved over the course of the simulation rather than representing a transient configuration sampled only during the early stages.

Global and local structural stability were further evaluated using root-mean-square deviation (RMSD) analysis. The protein backbone exhibited a low mean RMSD of 0.960 ± 0.143 Å ($SE = 0.0035$ Å) relative to the starting covalent structure, indicating that the overall KRAS fold remained structurally stable throughout the simulation. The reconstructed ligand displayed a mean heavy-atom RMSD of 0.868 ± 0.125 Å ($SE = 0.0031$ Å), demonstrating that the ligand remained bound in a stable orientation within the switch-II pocket while undergoing only modest conformational fluctuations. Collectively, the stable protein backbone, persistent covalent bond geometry, and low ligand RMSD support sustained structural integrity of the reconstructed covalent KRAS G12C–ligand complex over the full production trajectory.

These observations contrast sharply with the behavior observed in the corresponding non-covalent simulations. Although the manually reconstructed ligand could initially be positioned in a productive pre-reactive orientation, the absence of covalent anchoring resulted in rapid relaxation away from the engineered configuration during equilibrium molecular dynamics. In contrast, introduction of the explicit covalent linkage stabilized the reconstructed geometry over the full nanosecond timescale while preserving both the covalent bond architecture and the original switch-II pocket binding mode.

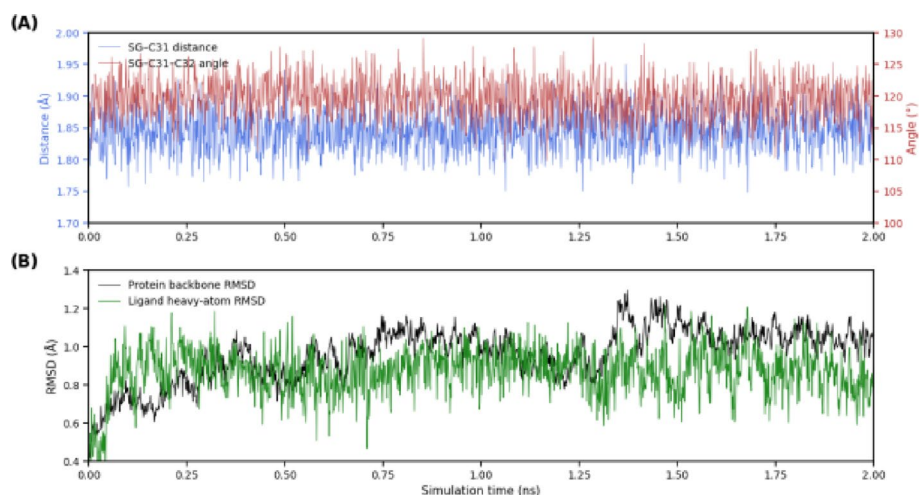


Fig. 11 Structural stability of the reconstructed covalent KRAS G12C–ligand complex during the ~2 ns production molecular dynamics simulation. **(A)** Time evolution of the covalent Cys12 SG–C31 bond distance (blue, left axis) and the SG–C31–C32 bond angle (red, right axis), demonstrating preservation of the engineered covalent geometry throughout the trajectory. **(B)** Time evolution of the protein backbone Ca root-mean-square deviation (RMSD, black) and ligand heavy-atom RMSD (green), indicating stable protein structure and retention of the ligand binding pose over the course of the simulation

Collectively, these findings demonstrate that covalent anchoring can stabilize a manually engineered reactive geometry that is not dynamically maintained under non-covalent conditions, providing mechanistic insight into the role of covalent bond formation in preserving productive KRAS G12C inhibitor conformations (Fig. 11).

3.7 Implications for covalent KRAS G12C inhibitor design

The present study provides several mechanistic insights directly relevant to the computational design of covalent KRAS G12C inhibitors. Across multiple independently evaluated ligand systems, including docking-derived scaffolds, manually reconstructed geometries, and benchmark covalent inhibitors, the results consistently demonstrate that reactive geometry cannot be reliably inferred from static docking poses alone. Although several ligands transiently achieved near-reactive sulfur–electrophile distances, these configurations frequently collapsed during molecular dynamics simulations and failed to preserve the coupled distance-and-angle requirements necessary for productive covalent engagement. These observations indicate that covalent feasibility should be evaluated as a dynamic property governed by persistence of reactive geometry under equilibrium conditions rather than by instantaneous docking proximity alone.

The comparative analyses performed in the present work further demonstrate that preservation of reactive geometry depends strongly on stabilization of the electrophilic exit vector relative to the Cys12 nucleophile. In both the manually reconstructed ligand system and the benchmark non-covalent Sotorasib simulations, the reactive configuration relaxed away from the crystallographic alignment despite initial near-reactive placement. In contrast, explicit covalent anchoring produced stable preservation of sulfur–carbon distance and attack geometry throughout molecular dynamics sampling. Collectively, these findings suggest that productive KRAS G12C inhibitor design requires simultaneous optimization of three coupled factors: (i) stable switch-II pocket occupancy, (ii) preservation of electrophilic trajectory toward Cys12, and (iii) dynamic stabilization of reactive geometry under thermal motion.

These observations have practical implications for future covalent drug discovery workflows targeting KRAS G12C and related oncogenic systems. Conventional docking-based screening strategies frequently prioritize binding affinity or short sulfur–electrophile distances as primary indicators of covalent feasibility. However, the present results demonstrate that geometrically accessible poses may remain dynamically unstable and therefore may not represent productive pre-reactive states. Incorporation of short-timescale molecular dynamics validation, coupled distance–angle analysis, and experimentally derived reaction-vector constraints may therefore improve prioritization of covalent inhibitor candidates before costly synthesis and biological testing. Such approaches could help reduce false-positive docking predictions and accelerate identification of scaffolds capable of maintaining productive reactive alignment.

More broadly, the mechanistic framework developed here may contribute to the rational design of next-generation covalent inhibitors by emphasizing dynamic preservation of reactive geometry as a central design objective rather than treating covalent bond formation as a purely proximity-driven event. Because KRAS G12C remains one of the most clinically important oncogenic covalent drug targets in precision oncology, improved understanding of the dynamic factors governing covalent stabilization may support development of more reliable computational screening strategies for future targeted cancer therapeutics.

The present findings are also relevant to the design of non-covalent KRAS inhibitors. Although reversible inhibitors do not require covalent bond formation, they must still stabilize productive binding orientations within the highly dynamic Switch-II pocket. The observed loss of near-reactive geometries during non-covalent molecular dynamics simulations demonstrates that static docking poses alone may overestimate binding competence if dynamic stability is not considered. Consequently, incorporation of short-timescale molecular dynamics simulations and geometric persistence analyses into early-stage virtual screening may improve identification of non-covalent scaffolds capable of maintaining favorable binding conformations. Such strategies may enhance the rational development of both reversible and covalent inhibitors targeting KRAS mutations by emphasizing dynamic conformational stability in addition to binding affinity.

3.8 Scope and limitations

The present study was designed as a mechanistic computational analysis of reactive geometry preservation in KRAS G12C covalent inhibitor systems rather than a complete drug-development pipeline. The primary objective was to investigate how sulfur–electrophile alignment evolves dynamically across docking-derived, manually reconstructed, and covalently stabilized ligand systems using molecular dynamics simulations. Consequently, the conclusions of this work should be interpreted within the context of covalent geometry persistence and stabilization rather than as direct predictions of biological potency or clinical efficacy.

Several limitations should therefore be considered. First, the simulations were conducted on relatively short molecular dynamics timescales, which may not fully capture slower conformational rearrangements or rare reactive events known to influence KRAS G12C binding behavior [1, 10]. Although consistent trends were observed across multiple independently analyzed systems, longer-timescale and enhanced-sampling simulations would provide more complete characterization of reactive-state populations.

Second, the present analysis focused primarily on geometric and dynamic descriptors of covalent feasibility, including sulfur–electrophile distance and attack-angle preservation. Explicit quantum mechanical treatment of transition states, activation barriers, and electronic reactivity was not performed. Accordingly, the simulations evaluate dynamic preservation of near-reactive configurations rather than direct covalent reaction kinetics or bond-formation probabilities [10, 11].

In addition, the manually reconstructed ligand investigated here was developed as a mechanistic probe to test stabilization of engineered reactive geometry. The scaffold was not optimized for medicinal chemistry properties such as synthetic accessibility, selectivity, or pharmacokinetic behavior. Therefore, the reconstructed system should not be interpreted as a validated therapeutic candidate, but rather as a computational model used to investigate covalent stabilization principles.

Finally, force-field-based molecular dynamics simulations inherently approximate electronic polarization and reactive chemistry. Although widely validated AMBER- and GAFF-based approaches were employed [15, 21], future QM/MM studies and experimental validation would further strengthen the mechanistic conclusions presented here.

Despite these limitations, the consistent agreement between docking analysis, non-covalent simulations, manually reconstructed systems, and benchmark covalent simulations supports the central conclusion of this work: productive KRAS G12C covalent engagement depends not only on achieving reactive proximity, but also on dynamically preserving reactive geometry under equilibrium conditions. Within this scope, the present study provides a mechanistic computational framework that may assist future development and evaluation of covalent KRAS G12C inhibitors.

The present molecular dynamics simulations were designed to evaluate the early-time dynamic preservation or loss of productive sulfur–electrophile geometry rather than to exhaustively sample the complete conformational landscape of the KRAS G12C–ligand complexes. We acknowledge that longer molecular dynamics trajectories would provide more extensive conformational sampling and could further characterize slower structural rearrangements. However, the principal mechanistic behaviors examined in this study—including the rapid loss of near-reactive geometry in the non-covalent systems and the stable preservation of chemically consistent sulfur–carbon geometry in the covalent complexes—were consistently observed within the available simulated timescale and remained supported by complementary structural metrics, including sulfur–carbon distance, attack-angle evolution, backbone RMSD, radius of gyration, and residue flexibility analyses. Accordingly, the conclusions of the present work are limited to the mechanistic behavior observed within the investigated timescale, while longer-timescale simulations represent an important direction for future investigation.

4 Conclusions

This study demonstrates that covalent feasibility in KRAS G12C systems depends not only on achieving reactive proximity, but also on dynamically preserving reactive geometry under equilibrium conditions. Across docking-derived ligands, manually reconstructed systems, and benchmark non-covalent simulations, near-reactive configurations frequently relaxed during molecular dynamics simulations despite favorable initial placement within the switch-II pocket. In contrast, explicitly covalent systems

exhibited stable preservation of sulfur–carbon distance and attack geometry throughout production dynamics.

The comparative analyses further show that reactive geometry should be evaluated as a coupled dynamic property involving both electrophilic trajectory and attack-angle preservation rather than as a static docking criterion alone. Importantly, manually reconstructed systems demonstrated that restoration of a near-reactive geometry is insufficient unless the scaffold can stabilize that configuration dynamically. The observed stabilization of previously unstable geometries following covalent anchoring further suggests that covalent attachment itself plays a major role in maintaining productive alignment within the KRAS G12C binding environment.

Collectively, these findings highlight limitations of conventional affinity-driven covalent docking approaches and support incorporation of geometry-aware molecular dynamics validation into future KRAS G12C inhibitor design workflows. Within this framework, the present study provides a mechanistic computational strategy for evaluating reactive-state persistence that may assist future development and prioritization of covalent targeted cancer therapeutics.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s44345-026-00072-w>.

Supplementary Material 1.

Author contributions

This is a single author paper. Saroj Khanal conceived the study, designed the computational workflow, performed molecular docking and molecular dynamics simulations, analyzed the data, interpreted the results, prepared the figures and tables, and wrote and revised the manuscript.

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Data availability

The datasets generated and analyzed during this study, including ligand structures, docking models, molecular dynamics trajectories, topology files, and supporting analysis data, are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable. This study did not involve human participants, human data, human tissue, or animals. Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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